

WAFRA, Kuwait

WMO: 405920

Lat: 28.567N

Lon: 48.067E

Elev: 107

StdP: 100.04

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.4	(c) 5.9	(d) -9.6	(e) 1.7	(f) 13.5	(g) -7.6	(h) 2.0	(i) 15.8	(j) 8.4	(k) 15.4	(l) 7.8	(m) 16.0	(n) 2.3	(o) 280	(p) 0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	47.7	21.4	46.7	21.3	45.7	21.1	28.3	33.5	26.7	34.8	25.0	34.8	4.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	23.0	32.1	24.5	19.8	31.5	22.2	17.1	30.1	92.4	33.5	84.7	34.9	76.8	34.8	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.6	(b) 7.6	(c) 6.8	(e) 1.3	(f) 49.8	(g) 1.3	(h) 0.6	(i) 0.4	(j) 50.2	(k) -0.4	(l) 50.6	(m) -1.1	(n) 50.9	(o) -2.0	(p) 51.4		
(a) 8.6	(b) 7.6	(c) 6.8	(e) -0.9	(f) 29.7	(g) 1.9	(h) 1.1	(i) -2.3	(j) 30.5	(k) -3.4	(l) 31.1	(m) -4.5	(n) 31.7	(o) -5.8	(p) 32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	13.9	16.4	21.0	26.3	32.8	37.3	38.9	38.4	34.7	29.4	20.7	15.7		
	DBStd	9.37	3.22	3.42	3.53	3.17	2.71	2.01	1.57	1.52	2.04	2.63	3.40	2.85		
	HDD10.0	8	6	1	0	0	0	0	0	0	0	0	0	1		
	HDD18.3	326	141	69	12	0	0	0	0	0	0	0	14	89		
	CDD10.0	6279	127	182	342	489	707	818	896	879	742	600	321	177		
	CDD18.3	3556	3	17	95	239	449	568	638	621	492	342	85	7		
	CDH23.3	55375	23	134	872	2643	6829	9906	11422	10840	7805	4162	686	53		
(13)	CDH26.7	39265	1	33	344	1354	4571	7512	8942	8361	5484	2424	236	4		
(14)	Wind	WSAvg	3.0	2.9	3.1	3.1	3.0	3.1	3.8	3.1	2.8	2.4	2.5	2.7		
(15) Precipitation	PrecAvg	182	35	23	31	19	5	1	1	1	0	4	32	30		
	PrecMax	298	103	81	97	52	20	2	3	2	1	26	182	90		
	PrecMin	82	1	1	1	1	0	0	0	0	0	0	1	1		
	PrecStd	61	26	20	24	15	5	1	1	0	0	6	39	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	30.7	38.2	40.5	45.2	48.5	49.2	48.7	46.1	42.1	34.9	27.3		
		MCWB	14.3	15.2	17.0	18.7	20.4	21.2	21.4	21.9	21.8	19.8	17.1	14.5		
	2%	DB	23.5	26.7	33.2	37.6	43.6	47.2	47.9	47.3	44.2	40.0	31.5	24.7		
		MCWB	13.9	14.1	15.8	17.8	20.0	20.9	21.6	21.8	21.0	19.4	16.7	14.9		
	5%	DB	21.5	24.7	30.1	35.5	41.7	45.8	46.8	46.2	43.1	38.2	29.3	23.0		
		MCWB	13.1	13.4	15.3	17.1	19.5	20.3	21.2	21.7	20.6	19.1	16.4	14.3		
	10%	DB	19.7	22.8	27.7	33.3	40.2	44.6	45.7	45.3	41.8	36.5	27.3	21.4		
		MCWB	12.4	12.9	14.8	16.8	19.2	20.0	21.0	21.5	20.3	19.0	16.1	13.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	17.7	20.5	21.0	23.7	25.2	28.6	29.7	29.0	26.6	22.6	19.2		
		MCDB	20.6	22.6	26.2	28.7	33.7	36.7	33.6	34.5	33.2	31.0	25.3	22.0		
	2%	WB	16.5	16.7	19.2	20.3	22.2	22.5	26.0	28.7	27.7	25.0	20.9	17.9		
		MCDB	19.9	21.8	24.2	29.0	34.4	42.8	35.4	33.9	32.9	28.9	25.2	20.9		
	5%	WB	15.0	15.7	17.9	19.5	21.3	21.4	23.9	27.4	25.8	23.8	19.4	16.6		
		MCDB	18.5	20.8	23.6	28.9	35.3	43.4	38.8	34.8	34.1	29.0	24.4	20.3		
	10%	WB	13.7	14.5	16.8	18.7	20.6	20.6	22.3	25.4	23.6	22.6	18.3	15.1		
		MCDB	18.1	20.2	24.0	28.5	35.5	42.9	42.8	37.5	35.1	28.6	23.7	19.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.2	13.1	13.0	13.9	14.7	14.4	15.0	15.1	14.1	11.7	11.4		
		MCDBR	12.9	14.3	16.3	15.7	15.9	16.0	15.5	15.8	15.9	15.9	14.2	12.7		
		MCWBR	5.9	5.7	5.8	4.8	4.7	5.1	4.9	5.5	5.6	5.8	5.8	5.7		
	5% WB	MCDBR	10.6	11.3	11.5	12.4	14.2	16.1	16.0	15.2	14.4	12.9	10.5	9.7		
		MCWBR	6.1	5.5	5.5	4.2	4.7	5.7	6.3	7.0	6.4	5.8	5.2	5.6		
(40) Clear-Sky Solar Irradiance	taub		0.508	0.559	0.614	0.701	0.704	0.666	0.673	0.626	0.572	0.558	0.543	0.504		
	taud		1.772	1.670	1.559	1.461	1.462	1.496	1.509	1.595	1.683	1.710	1.735	1.792		
	Ebn at Noon		697	701	696	655	656	678	671	699	720	691	657	675		
	Edh at Noon		190	228	270	307	308	296	291	265	235	216	196	179		
(44) All-Sky Solar Radiation	RadAvg		3.48	4.27	5.30	5.79	6.62	7.37	7.04	6.73	6.06	4.88	3.60	3.18		
	RadStd		0.23	0.27	0.27	0.31	0.29	0.29	0.19	0.14	0.20	0.21	0.27	0.28		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.35	N/A	N/A	+0.52	N/A	N/A	N/A	+129	+136

Nomenclature: See separate page